

DATA ANALYTICS & BUSINESS INTELLIGENCE PROJECT

# Amazon Products Sales Analysis (2025)

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PROJECT BY

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NTI Final Project | Amazon Product Data | 2025

# Project Overview

This project analyzes Amazon electronics product data collected in 2025 to identify patterns in pricing, discounts, ratings, customer engagement, product positioning and category performance.

- Product pricing
- Discounts
- Customer ratings
- Customer engagement
- Product positioning
- Product / category performance

**42,675**

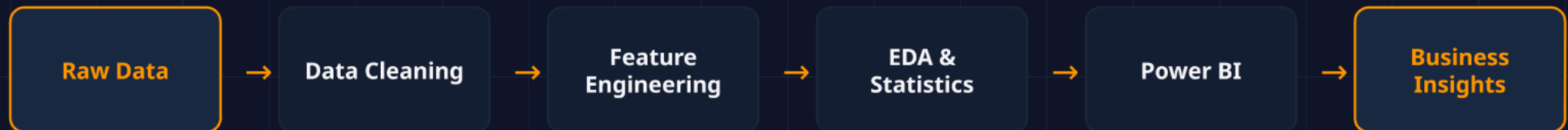
Products / Records

**16**

Features /  
Columns

Data Source: Amazon Product Data  
Year:  
**2025**

## ANALYTICAL WORKFLOW



# Business Questions

## 01 Pricing

How does pricing vary across products and price segments?

## 02 Discounts

How are discounts distributed across products, and what patterns can we observe?

## 03 Customer Engagement

What patterns exist between product ratings, reviews and customer engagement?

## 04 Product Performance

Which products or categories stand out as opportunities or areas requiring attention?

# Data Cleaning & Preprocessing

From raw data to analysis-ready data

## Missing Values

- Missing-value analysis performed
- Missing percentages calculated
- Columns with substantial gaps identified
- Removed sustainability\_badges (extreme missing %)

## Missing Value Handling

- Categorical / text gaps filled with "Not Available"
- buy\_box\_availability
- delivery\_details
- product\_url
- bought\_in\_last\_month

## Numeric Cleaning

- rating
- number\_of\_reviews
- current/discounted\_price
- "4.6 out of 5 stars" → 4.6
- "35,882" → 35882

## Data Validation

- Missing values
- Duplicate rows
- Invalid ratings
- Negative reviews
- Negative prices

Raw Data



Missing Value Analysis



Remove High-Missing Column



Clean Text & Numeric



Handle Missing Values



Validation



Clean Dataset

# Data Quality Validation

0

Duplicate Rows

0

Invalid Ratings

0

Negative Reviews

0

Negative Prices

**The dataset passed the main data-quality validation checks.**

# Feature Engineering

New analytical features created from the cleaned data

## Discount Percentage

Discount % =  $((\text{Listed Price} - \text{Current Price}) / \text{Listed Price}) \times 100$

\$159 → \$89.68 **43.6% Discount**

\$15.99 → \$9.99 **37.5% Discount**

\$349 → \$314 **10.0% Discount**

## Price Bucket

**Low** 0 – 50

**Medium** 50 – 200

**High** > 200

*Price buckets make it easier to compare products across different pricing levels.*

**Raw Prices**



**Discount %**



**Price Buckets**

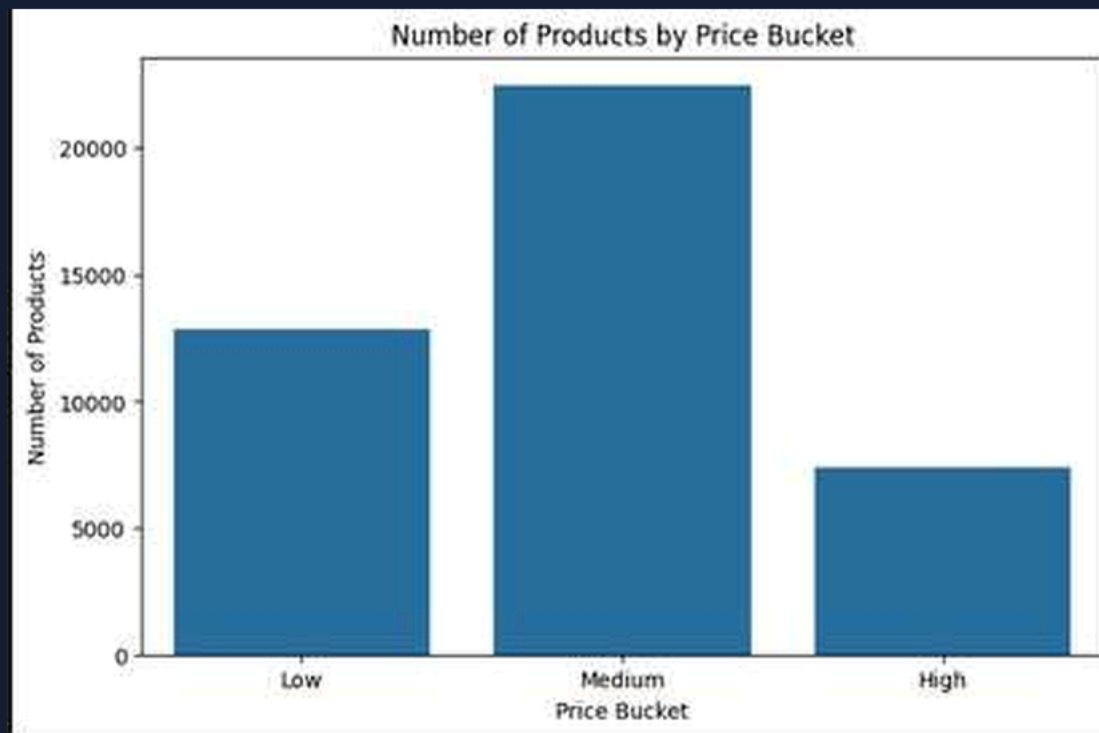


**Business Analysis**

# Python EDA — Part 1

Pricing, Product Distribution & Ratings

## Price Bucket Distribution

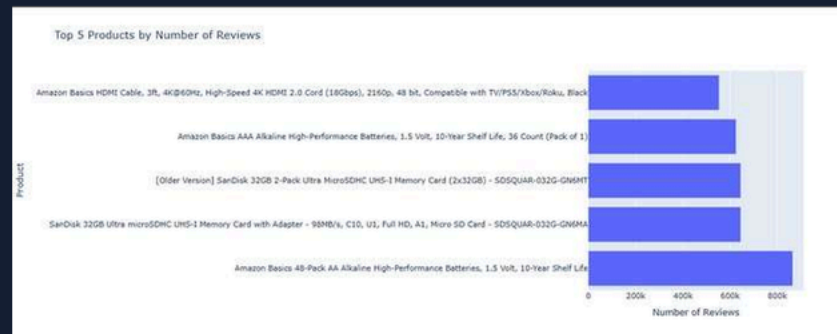


Key Insight: Medium-priced products dominate the catalogue, followed by Low, then High.

# Python EDA — Part 2

## Discounts, Reviews & Customer Engagement

### Top Products by Reviews



Key Insight: Amazon Basics & SanDisk accessories lead on review volume.

### Avg. Engagement by Discount Group

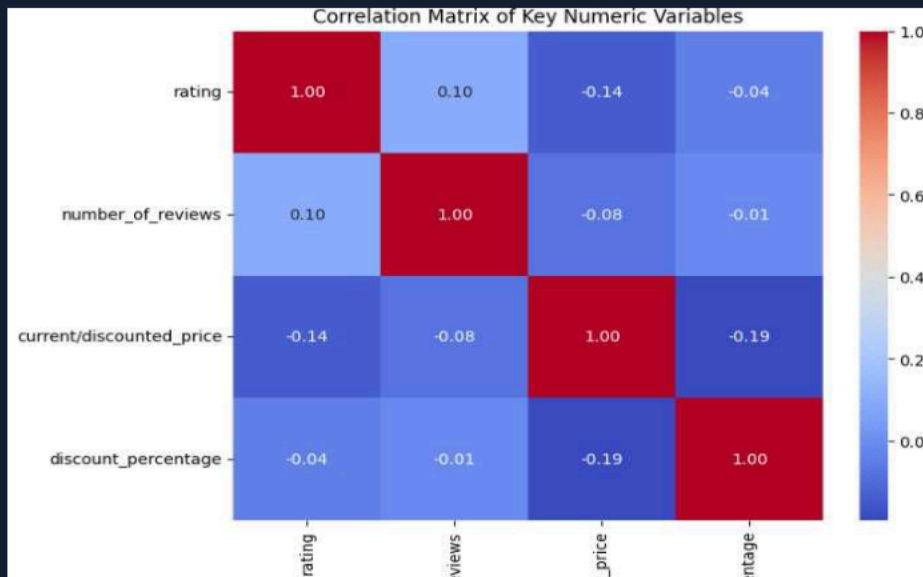


Key Insight: Average reviews are close for low- vs high-discount groups.



# Statistical Analysis

Correlation Matrix of Key Numeric Variables



Customer satisfaction has a weak positive relationship with customer engagement — higher-rated products do not necessarily attract significantly more reviews.

**-0.508**

T-statistic

**0.612**

P-value

**Not Statistically Significant (p = 0.612 > 0.05)**

## Interpretation

The difference in customer engagement between the compared discount groups was not statistically significant. There is insufficient statistical evidence to conclude that discount level is associated with a significant difference in customer engagement.

# From Python Analysis to Power BI

## PYTHON

Analyze & Discover

- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis
- Statistics
- Relationship Analysis

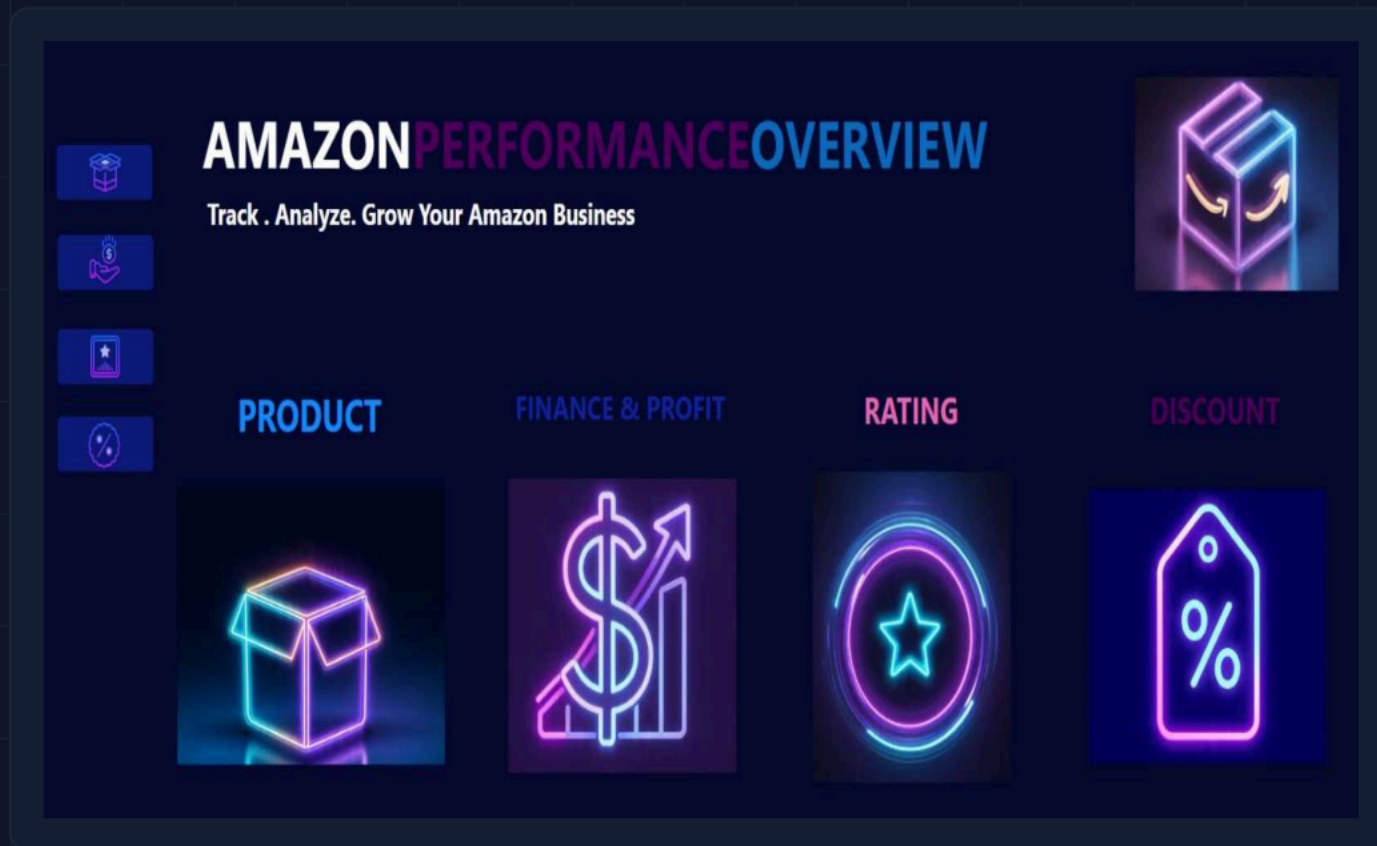


## POWER BI

Visualize & Decide

- Interactive Dashboards
- KPIs
- Product & Category Comparison
- Business Insights
- Decision Support

# Power BI Dashboard — Amazon Performance Overview



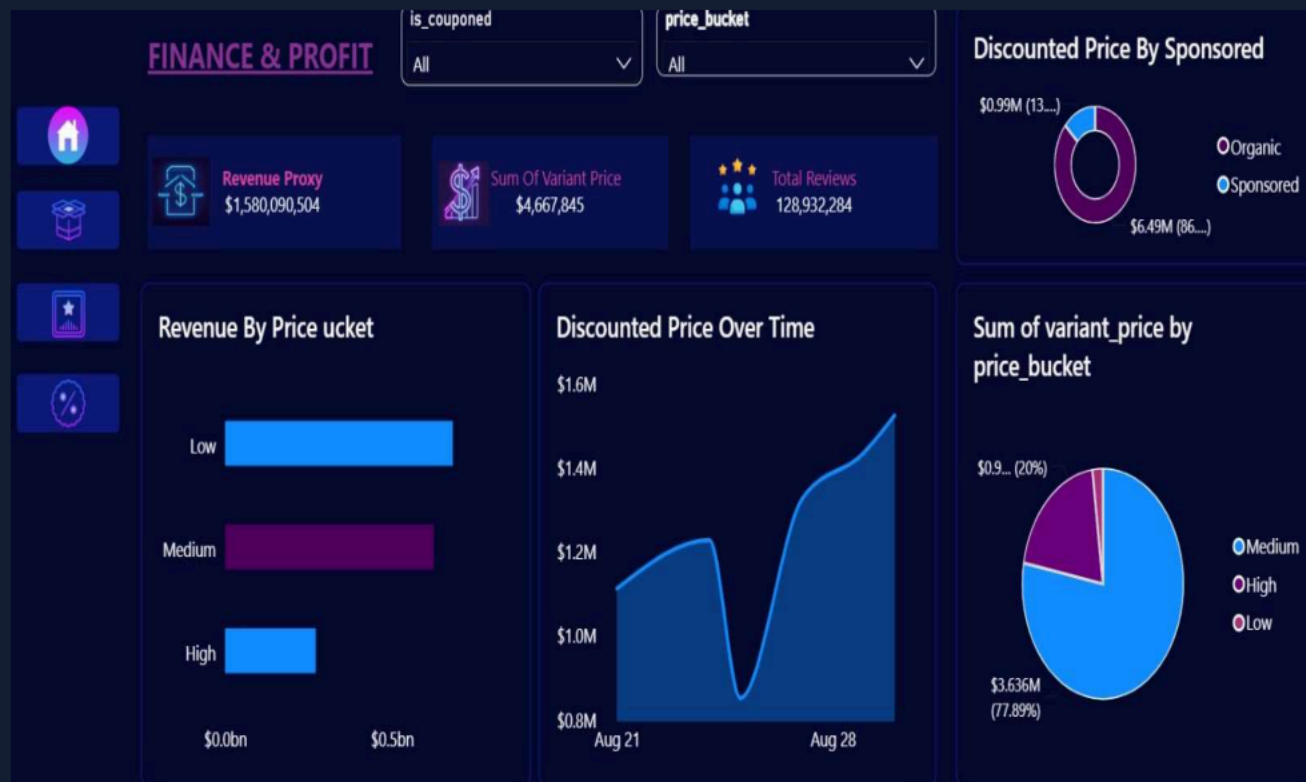
## Dashboard Focus

- Landing page navigating four analysis areas
- Product · Finance & Profit · Rating · Discount
- Single entry point for the full product story

## Key Insights

- [INSERT ACTUAL INSIGHT]
- [INSERT ACTUAL INSIGHT]
- [INSERT ACTUAL INSIGHT]

# Power BI Dashboard — Finance & Profit



## What Does This Dashboard Show?

- KPIs: Revenue Proxy \$1,580,090,504 · Sum of Variant Price \$4,667,845 · Total Reviews 128,932,284
- Revenue by price bucket and discounted price over time
- Filters for is\_couponed and price\_bucket; Organic vs Sponsored split

## Key Insights

- Medium price bucket holds 77.89% (\$3.636M) of variant price value.
- Sponsored listings account for \$6.49M (≈86%) of discounted price.
- [INSERT ACTUAL INSIGHT]

# Power BI Dashboard — Rating



## Dashboard Insights

- KPIs: Sum of Rating 187,848.70 · Avg Rating 4.4 · Total Rating Segment 42,675
- Rating distribution, rating vs discounted price, rating by brand
- Filters by rating\_segment and brand

## Key Insights

- Ratings concentrate between 4.0 and 5.0, peaking near 4.5.
- High rating segment covers roughly 85% of products (3.09K vs 0.5K).
- ASURION, HP and Duracell lead on rating volume by brand.

# Key Findings

**01**

## Pricing

Medium-priced products dominate the catalogue, with Low and High segments trailing.

**02**

## Discounts

Discount levels vary widely; high-discount products do not stand apart on engagement.

**03**

## Customer Engagement

Review volume is highly concentrated in a small set of accessory products.

**04**

## Customer Satisfaction

Average rating is 4.4; ratings cluster tightly between 4.0 and 5.0.

**05**

## Product / Category Performance

[INSERT ACTUAL FINDING]

Correlations between rating, reviews, price and discount are weak — no single driver explains engagement.

# Business Recommendations

## Pricing Strategy

Monitor product price segments and identify opportunities across Low, Medium and High price products.

## Discount Strategy

Do not assume higher discounts automatically generate higher engagement — the test found no significant difference.

## Product Monitoring

Monitor products with strong ratings and customer engagement.

## Opportunity Identification

Identify products and categories combining strong satisfaction with strong engagement.

## Dashboard Monitoring

Use Power BI dashboards for continuous monitoring and business decision support.

# Conclusion



**“Turning raw product data into clear, data-driven business decisions.”**

# Thank You

Questions?